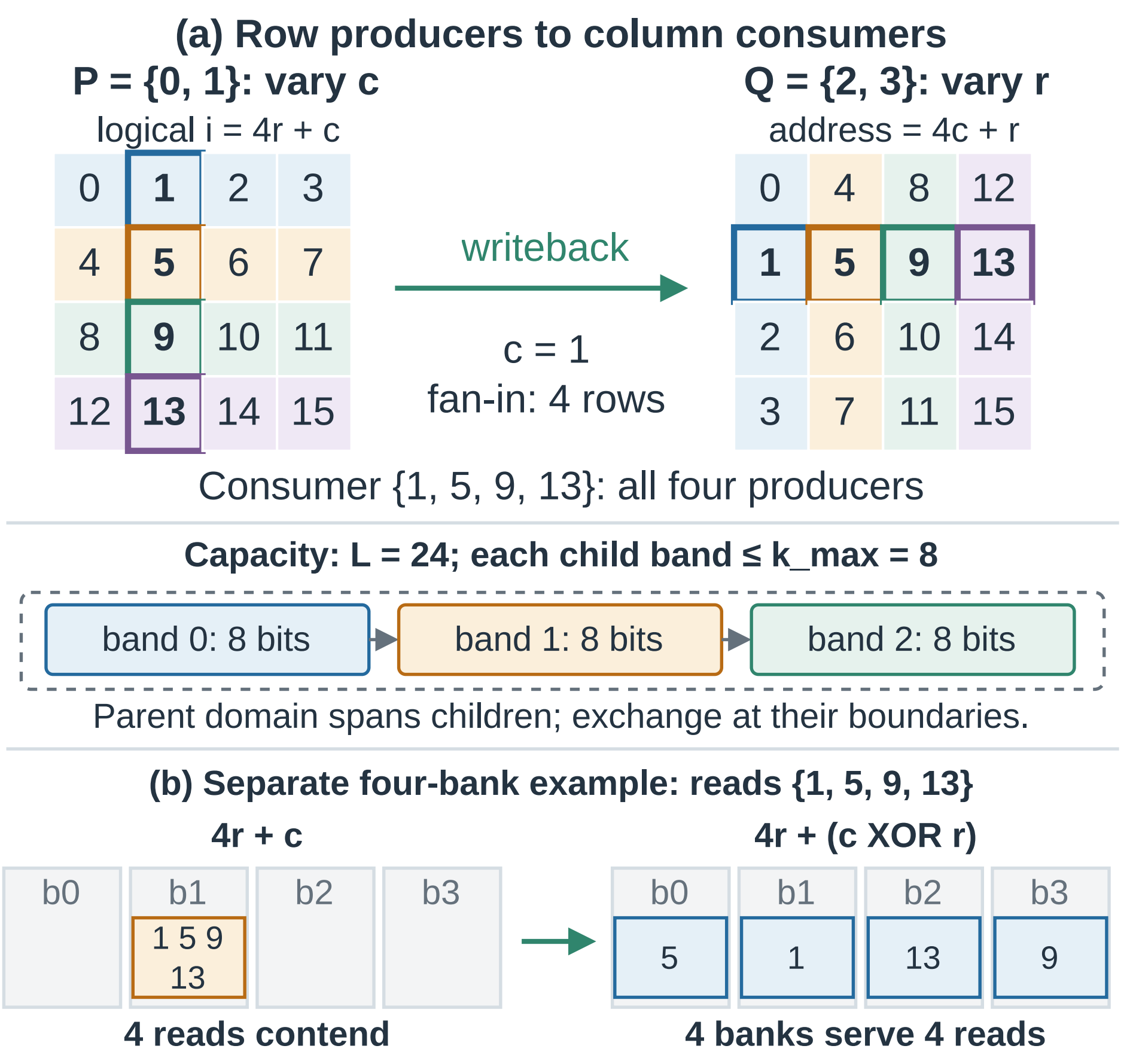
(a) Row producers to column consumers
P = {0, 1}: vary c
logical i = 4r + c
0 1 2 3
4 5 6 7
8 9 10 11
12 13 14 15
writeback
c = 1
fan-in: 4 rows
Q = {2, 3}: vary r
address = 4c + r
0 4 8 12
1 5 9 13
2 6 10 14
3 7 11 15
Consumer {1, 5, 9, 13}: all four producers
Capacity: L = 24; each child band ≤ k_max = 8
band 0: 8 bits
band 1: 8 bits
band 2: 8 bits
Parent domain spans children; exchange at their boundaries.
(b) Separate four-bank example: reads {1, 5, 9, 13}
4r + c
b0
b1
1 5 9
13
b2
b3
4 reads contend
4r + (c XOR r)
b0
5
b1
1
b2
13
b3
9
4 banks serve 4 reads